

About me

My expertise lies in Spatial AI systems: designing customer-facing agentic front-ends grounded in a 3D-vision backend spanning camera calibration, localization, reconstruction, and Structure-from-Motion.

Work Experience

Aug, 2024 – now

Applied Scientist II, Ring AI, Amazon.

- Developed [Ask Ring](#), a spatial-aware agentic system for Ring Cameras.
 - Led Intent Routing, Content Moderation, and primary contributor to Video Query Agent.
 - A robust LiDAR-Camera calibration method, friendly to inexperienced data-collection teams. [\[Publication-7\]](#)
 - Indoor & Outdoor Localization and Reconstruction. [\[Publication-1\]](#)
- Developed the Ring Humanoid Robot (Unitree-based): world-model training for localization, navigation, and operation.
- Contributed to Ring Detection models, power- and memory-optimized detectors running on-device on Ring Camera SoCs.
- Contributed to Ring Smart Video Descriptions, a light-weight real-time VLM narrating camera events.
- Co-designed storage tiering and training-pipeline integration for Ring Parallel Computing Cluster on AWS Slurm.

May – Aug, 2024

Research Scientist Intern, BAIR, Google.

Foundation depth model based Structure-from-Motion for Neural Rendering and AR/VR. [\[Publication-2\]](#)

2021 & 2022

Applied Scientist Intern, Amazon Device AI.

Improve the LiDAR generated depthmap quality with epipolar geometry. [\[Publication-8\]](#)

Education

2017 – 2024:

PhD, Computer Science, Michigan State University, East Lansing, U.S.

Dissertation: Structure and Motion Estimation from Dense Depth and Correspondence

Advisor: Prof. Xiaoming Liu

2013 – 2017:

Bachelor of Engineering, Electrical and Electronics, Southeast University, Nanjing, China.

First-Authored Publications

Localization, Reconstruction, and Structure-from-Motion:

(Under Review)

[1] On Self-Improving Structure-from-Motion: A Generalizable Multi-View Pose Estimator [\[Website\]](#).

[Shengjie Zhu](#) and others

3DV'26

[2] Marginalized Bundle Adjustment: Multi-View Camera Pose from MonoDepth [\[PDF, Website\]](#).

[Shengjie Zhu](#), Ahmed Abdelkader, Mark J. Matthews, Xiaoming Liu, and Wen-Sheng Chu

ECCV'24

[3] Revisit Self-Supervision with Local Structure-from-Motion [\[PDF, Website\]](#).

[Shengjie Zhu](#), Xiaoming Liu

CVPR'23

[4] LightedDepth: Video Depth Estimation in light of Limited Inference View Angles [\[PDF, Website\]](#).

[Shengjie Zhu](#), Xiaoming Liu

CVPR'23

[5] PMatch: Paired Masked Image Modeling for Dense Geometric Matching [\[PDF, Website\]](#).

[Shengjie Zhu](#), Xiaoming Liu

CVPR'20

[6] The Edge of Depth: Explicit Constraints between Segmentation and Depth [\[PDF, Website\]](#).

[Shengjie Zhu](#), Garrick Brazil, Xiaoming Liu

Sensor Calibration:

(Under Review)

[7] Camera LiDAR Calibration in the Eye of LiDAR [\[Website\]](#).

[Shengjie Zhu](#) and others

ECCV'24

[8] Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry [\[PDF, Website\]](#).

[Shengjie Zhu](#)*, Girish Chandar Ganesan*, Abhinav Kurmur, Xiaoming Liu

NeurIPS'23

[9] Tame a Wild Camera: In-the-Wild Monocular Camera Calibration [\[PDF, Website\]](#).

[Shengjie Zhu](#), Abhinav Kurmur, Masa Hu, Xiaoming Liu

Computer Skills

Language

Python, C++, CUDA, Matlab, PyTorch, Tensorflow, CuPy, Numba

Talk

Aug. 08, 2023

3D Perception from Two Views, Google Pixel Biometrics Seminar.

Feb. 05, 2024

Structure-from-Motion Meets Self-supervised Learning, CMU VASC Seminar. [\[Link\]](#)

Jul. 09, 2024

Structure-from-Motion from Dense Learning, Google Pixel Biometrics Seminar.

Mentored Students

[Girish Chandar Ganesan](#) (MSU), [Qucheng Peng](#) (UCF), [Ziyao Wang](#) (UMD)

Academic Service

Reviewer for CVPR, ECCV, ICCV, NeurIPS, 3DV, BMVC, ACL ARR, Pattern Recognition, IEEE T-CSVT, IEEE T-MM, JVCIR, DSP.

Other Publications

CVPR'26 **[10] From 3D Pose to Prose: Biomechanics-Grounded Vision–Language Coaching** [\[PDF, Website\]](#).
Yuyang Ji, Yixuan Shen, [Shengjie Zhu](#), Yu Kong, Feng Liu